

Graphical Visualization Of The Complex Roots Of Parabolas

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1 Abstract

Quadratic functions, $f(x) = ax^2 + bx + c$, which have only complex roots, do not intersect the x-axis. In Figure 1, $f(x)$ is plotted along with its complex roots, which lie on the axis of symmetry of the parabola, $x = \frac{-b}{2a}$.

In this article, the y-axis will express both the output space for real functions as well as the imaginary component of complex numbers. The complex roots of $f(x)$ can be visualized by the following methodology:

1. Reflect function $f(x)$ about the horizontal tangent line ($y = \frac{-b^2}{4a} + c$) at the vertex of the parabola, resulting in function $h(x)$. Figure 2.
2. Function $h(x)$ will intersect the x-axis at two points.
3. Form a circle centered at the point $(-\frac{b}{2a}, 0)$, whose radius extends to the x-axis intersections of $h(x)$. ($Radius = \frac{\sqrt{-b^2+4ac}}{2a}$)
4. The resulting circle, C, will intersect the vertical axis of symmetry $x = \frac{-b}{2a}$ of $f(x)$, at its complex roots.

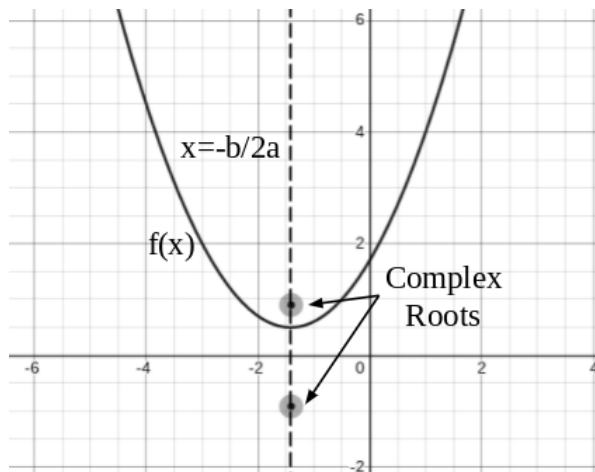


Figure 1: The quadratic function $f(x)$ with visualized complex roots.

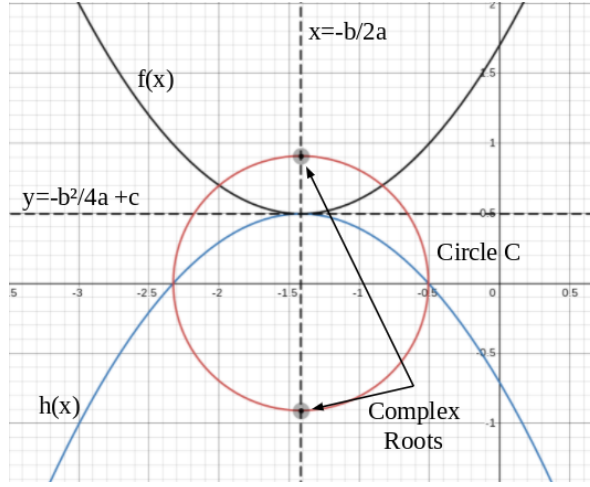


Figure 2: $f(x)$, its reflection $h(x)$, and circle C .

2 Explanation

Provided by the quadratic formula, it can be seen that the complex roots of a quadratic function $f(x) = ax^2 + bx + c$ fall upon the axis of symmetry of the parabola $x = \frac{-b}{2a}$. (Notice the imaginary y-component of these complex roots is the radius of circle C)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{Roots} = \frac{-b}{2a} \pm i \frac{\sqrt{-(b^2 - 4ac)}}{2a}$$

Equation 1.

The graphical technique for visualization preserves the axis-of-symmetry of $f(x)$, upon which the complex roots fall, by reflecting the parabola about a horizontal line. This horizontal line intersects the vertex of the parabola;

$$y = a \left(\frac{-b}{2a} \right)^2 + b \left(\frac{-b}{2a} \right) + c$$

$$y = \frac{-b^2}{4a} + c$$

The reflected function, $h(x)$, is guaranteed to intersect the x-axis. This is a consequence of the fact that the horizontal reflection line constitutes the border to the parabola's range limitations. Thus the newly reflected function's range will extend past either the $\left\{ \frac{-b^2}{4a} + c \leq y < \infty \right\}$ or $\left\{ -\infty < y \leq \frac{-b^2}{4a} + c \right\}$ ranges (depending on the sign of the a coefficient), to eventually intersect the x-axis.

The equation for the reflected parabola, $h(x)$, is evaluated as follows:

- For *any* function $f(x)$, the reflection across a horizontal line of $y = k$ is equal to $-f(x) + 2k$.¹
- In the case of the parabola, $k = \frac{-b^2}{4a} + c$.

¹The reflection about the horizontal $y = k$ line can be first thought of as a reflection about the x axis, producing the term $-f(x)$. Next, a vertical shift up to the line of reflection, followed by an additional shift by k to a point equidistant above the line, is expressed as the $+2k$ term.

- The reflected parabola, $h(x)$, is:

$$h(x) = -(ax^2 + bx + c) + 2\left(\frac{-b^2}{4a} + c\right)$$

$$h(x) = -ax^2 - bx + c - \frac{b^2}{2a}$$

Utilizing the quadratic formula, one can evaluate the roots of the reflected parabola $h(x)$ as such;

$$x = \frac{b \pm \sqrt{b^2 - 4 \cdot -a \cdot \left(c - \frac{b^2}{2a}\right)}}{-2a}$$

$$x = \frac{-b \pm \sqrt{b^2 + (4ac - 2b^2)}}{2a}$$

$$x = \frac{-b \pm \sqrt{-b^2 + 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{-(b^2 - 4ac)}}{2a}$$

Equation 2.

Observe that the complex roots of the original parabola (See Equation 1), are identical to the roots of $h(x)$ (Equation 2), except for the presence of the imaginary unit i .

Thus, if the roots of $h(x)$ have their x-components 'exchanged' as their new y-components, and if such resulting points have a 'new' x-component of $-\frac{b}{2a}$, they will represent the complex roots of $f(x)$. (Equation 1)

In order to accomplish this 'exchange', circle C must be constructed with a center at $(-\frac{b}{2a}, 0)$, and whose radius intersects the two roots of $h(x)$. The complex roots of the *original* quadratic function $f(x)$ will then be the points of circle C intersecting the axis of symmetry.

The equation for Circle C , some manipulation to the $x^2 + y^2 = r^2$ form, is defined by the horizontal translation to the $x = -\frac{b}{2a}$ line, and has a radius equal to the distance between the root and the line of symmetry;

$$\left(x - \frac{-b}{2a}\right)^2 + y^2 = \left(\frac{-b}{2a} - \frac{-b \pm \sqrt{-(b^2 - 4ac)}}{2a}\right)^2$$

$$\left(x + \frac{b}{2a}\right)^2 + y^2 = \left(\frac{\pm \sqrt{-(b^2 - 4ac)}}{2a}\right)^2$$

The intersection between Circle C and the axis of symmetry are found by substitution of x with $-\frac{b}{2a}$;

$$\left(-\frac{b}{2a} + \frac{b}{2a}\right)^2 + y^2 = \left(\frac{\pm \sqrt{-(b^2 - 4ac)}}{2a}\right)^2$$

$$y = \frac{\pm \sqrt{-(b^2 - 4ac)}}{2a}$$

The point $\left(\frac{-b}{2a}, \pm \frac{\sqrt{-(b^2 - 4ac)}}{2a}\right)$ in the x-y plane.

By the representation of 'y' as the imaginary component i , the roots of $f(x)$ are located: (Equation 1)

$$\frac{-b}{2a} \pm i \frac{\sqrt{-(b^2 - 4ac)}}{2a}$$